Chapter 11. The Seifert–van Kampen Theorem

Free products and free groups, the Seifert–van Kampen theorem, wedges of circles, attaching 2-cells

Contents

11.1 Direct sums of abelian groups **2**

11.2 Free products of groups **3**

11.3 Free groups **5**

11.4 The Seifert–van Kampen theorem **6**

11.5 The fundamental group of a wedge of circles **7**

11.6 Adjoining a two-cell **8**

11.7 The fundamental groups of the torus and the dunce cap **9**

11.1 Direct sums of abelian groups

Motivation. Direct sums, characterized by a universal property, model the free product to come.

Abelian groups are written additively. Subgroups G_α ($\alpha \in J$) of G **generate** G if every $x \in G$ is a finite sum $\sum x_\alpha$, $x_\alpha \in G_\alpha$ (almost all 0); G is their **direct sum**, $G = \bigoplus G_\alpha$, if this expression is unique. G is **free abelian** with **basis** $\{a_\alpha\}$ if each a_α generates an infinite cyclic G_α and $G = \bigoplus G_\alpha$.

Lemma 11.1 (67.1)

If $G = \bigoplus G_\alpha$, then G has the **extension property**: for every abelian group H and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique homomorphism $h : G \to H$ restricting to each h_α. Conversely, if the G_α generate G and the extension property holds, then $G = \bigoplus G_\alpha$.

Proof. If $G = \bigoplus G_\alpha$, $h(\sum x_\alpha) = \sum h_\alpha(x_\alpha)$ is well defined, a homomorphism, and forced. Conversely let $\sum x_\alpha = \sum y_\alpha$; for fixed β extend $h_\beta = \mathrm{id} : G_\beta \to G_\beta$, $h_\alpha = 0$ ($\alpha \neq \beta$), and apply the extension: $x_\beta = y_\beta$. ■

Corollary 11.2 (67.2)

If $G = G_1 \oplus G_2$, $G_1 = \bigoplus_{\alpha\in J} H_\alpha$ and $G_2 = \bigoplus_{\beta\in K} H_\beta$ with $J \cap K = \emptyset$, then $G = \bigoplus_{\gamma\in J\cup K} H_\gamma$.

Proof. The H_γ generate G; homomorphisms h_γ extend to G_1, G_2, then to G, uniquely as the H_γ generate. Apply Lemma 11.1. ■

Corollary 11.3 (67.3)

If $G = G_1 \oplus G_2$, then $G/G_2 \cong G_1$.

Proof. Extend $\mathrm{id} : G_1 \to G_1$ and $0 : G_2 \to G_1$ to $h : G \to G_1$, onto with kernel G_2. ■

Theorem 11.4 (67.4)

For any family $\{G_\alpha\}$ of abelian groups there are an abelian group G and monomorphisms $i_\alpha : G_\alpha \to G$ with $G = \bigoplus i_\alpha(G_\alpha)$ (the **external direct sum**).

Proof. Take the tuples in $\prod G_\alpha$ with finitely many nonzero entries, and $i_\beta(x)$ the tuple with x in coordinate β, 0 elsewhere; each element is uniquely a finite sum of such. ■

Lemma 11.5 (67.5)

If $i_\alpha : G_\alpha \to G$ are monomorphisms and $G = \bigoplus i_\alpha(G_\alpha)$, then for any abelian H and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique $h : G \to H$ with $h \circ i_\alpha = h_\alpha$.

Proof. Apply Lemma 11.1 to $h_\alpha \circ i_\alpha^{-1} : i_\alpha(G_\alpha) \to H$. ■

Theorem 11.6 (67.6, uniqueness of direct sums)

If $G = \bigoplus i_\alpha(G_\alpha)$ and $G' = \bigoplus i'_\alpha(G_\alpha)$ with monomorphisms i_α, i'_α, there is a unique isomorphism $\phi : G \to G'$ with $\phi \circ i_\alpha = i'_\alpha$.

Proof. Lemma 11.5 gives ϕ with $\phi i_\alpha = i'_\alpha$ and ψ with $\psi i'_\alpha = i_\alpha$. Then $\psi\phi i_\alpha = i_\alpha$, so $\psi\phi = \mathrm{id}$ by uniqueness ($H = G$, $h_\alpha = i_\alpha$); similarly $\phi\psi = \mathrm{id}$. ■

Lemma 11.7 (67.7)

Let $\{a_\alpha\}$ generate the abelian group G. Then G is free abelian with basis $\{a_\alpha\}$ iff for every abelian H and elements $y_\alpha \in H$ there is a homomorphism $h : G \to H$ with $h(a_\alpha) = y_\alpha$; such h is then unique.

Proof. Let $G_\alpha = \langle a_\alpha \rangle$. If the condition holds, $H = \mathbb{Z}$, $y_\alpha = 1$ shows $na_\alpha = 0 \Rightarrow n = 0$; homomorphisms $G_\alpha \to H$ correspond to choices of y_α, so the extension property holds and $G = \bigoplus G_\alpha$ (Lemma 11.1). Conversely, $na_\alpha \mapsto ny_\alpha$ defines h_α on the infinite cyclic G_α; extend by Lemma 11.1. Uniqueness: the a_α generate. ■

Theorem 11.8 (67.8)

If G is free abelian with basis $\{a_1, \ldots, a_n\}$, then n is determined by G.

Proof. $2G = \{2x\}$ is the set of $\sum n_i a_i$ with all n_i even, the kernel of the surjection $\sum n_i a_i \mapsto (n_i \bmod 2) \in (\mathbb{Z}/2)^n$. So $|G/2G| = 2^n$, which depends only on G. ■

n is the **rank** of G.

11.2 Free products of groups

Motivation. The free product combines groups with no relations; van Kampen's theorem produces it.

Let $\{G_\alpha\}$ be subgroups generating a group G with $G_\alpha \cap G_\beta = \{1\}$ for $\alpha \neq \beta$. A **word** is a finite sequence $(g_1, \ldots, g_n)$ in $\bigcup G_\alpha$, representing $g_1 \cdots g_n$; it is **reduced** if no $g_i = 1$ and no consecutive g_i, g_{i+1} lie in one G_α. Every element has a reduced word (merge adjacent letters of one group, delete 1's); 1 has the empty word. G is the **free product** $\prod^* G_\alpha$ if each element has only one reduced word. It suffices that no nonempty reduced word represent 1: if reduced $(g_1, \ldots, g_n)$, $(h_1, \ldots, h_m)$ represent the same element, $g_1 \cdots g_n h_m^{-1} \cdots h_1^{-1} = 1$; if g_n, h_m lie in different groups this word is reduced and nonempty, if in one group and different, merging gives one, and if $g_n = h_m$ cancel and induct.

Lemma 11.9 (68.1)

If $G = \prod^* G_\alpha$, then for every group H and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique homomorphism $h : G \to H$ restricting to each h_α.

Proof. For x with reduced word $(g_1, \ldots, g_n)$, $g_i \in G_{\alpha_i}$, set

$$h(x) = h_{\alpha_1}(g_1) \cdots h_{\alpha_n}(g_n),$$

well defined by uniqueness. The formula gives the same value on any word (merging or deleting 1's does not change the product), and concatenation represents xy, so $h(xy) = h(x)h(y)$. Uniqueness: the G_α generate. ■

Theorem 11.10 (68.2)

For any family $\{G_\alpha\}$ of groups there are a group G and monomorphisms $i_\alpha : G_\alpha \to G$ such that G is the free product of the subgroups $i_\alpha(G_\alpha)$ (the **external free product**).

Proof. Take the $G_\alpha - \{1\}$ disjoint, and W the set of reduced words in their union (with the empty word). For $g \in G_\alpha$ define $\pi_g : W \to W$: $\pi_1 = \mathrm{id}$; for $g \neq 1$, $w = (g_1, \dots, g_n)$, $\pi_g(w) = (g, g_1, \dots, g_n)$ if w is empty or $g_1 \notin G_\alpha$; $(gg_1, g_2, \dots, g_n)$ if $g_1 \in G_\alpha$, $gg_1 \neq 1$; $(g_2, \dots, g_n)$ if $gg_1 = 1$. By cases $\pi_g \pi_{g'} = \pi_{gg'}$ for $g, g' \in G_\alpha$ (both change only a first letter in G_α). So π_g is a bijection with inverse $\pi_{g^{-1}}$, and $i_\alpha(g) = \pi_g$ is a homomorphism into the permutations of W, injective as $\pi_g(\emptyset) = (g)$. Let $G = \langle \pi_g \rangle$. For a nonempty reduced word, $\pi_{g_1} \cdots \pi_{g_n}(\emptyset) = (g_1, \dots, g_n) \neq \emptyset$, so it does not represent 1; likewise $i_\alpha(G_\alpha) \cap i_\beta(G_\beta) = \{1\}$. ∎

Lemma 11.11 (68.3)

If $i_\alpha : G_\alpha \to G$ are monomorphisms and G is the free product of the $i_\alpha(G_\alpha)$, then for every group H and homomorphisms $h_\alpha : G_\alpha \to H$ there is a unique $h : G \to H$ with $h \circ i_\alpha = h_\alpha$.

Proof. Apply Lemma 11.9 to $h_\alpha \circ i_\alpha^{-1}$. ∎

Theorem 11.12 (68.4, uniqueness of free products)

Let $i_\alpha : G_\alpha \to G$ and $i'_\alpha : G_\alpha \to G'$ be monomorphisms whose images generate G and G', both with the extension property of Lemma 11.11. Then there is a unique isomorphism $\phi : G \to G'$ with $\phi \circ i_\alpha = i'_\alpha$.

Proof. As in Theorem 11.6: the extensions ϕ, ψ satisfy $\psi\phi i_\alpha = i_\alpha$, so $\psi\phi = \mathrm{id}$; likewise $\phi\psi = \mathrm{id}$. ∎

Lemma 11.13 (68.5)

If homomorphisms $i_\alpha : G_\alpha \to G$ satisfy the extension condition of Lemma 11.11, then each i_α is a monomorphism and G is the free product of the $i_\alpha(G_\alpha)$.

Proof. Extending $h_\beta = \mathrm{id} : G_\beta \to G_\beta$, h_α trivial ($\alpha \neq \beta$), gives $h \circ i_\beta = \mathrm{id}$, so i_β is injective. Let G', i'_α be the external free product (Theorem 11.10). Extensions give $\phi : G \to G'$, $\phi i_\alpha = i'_\alpha$, and $\psi : G' \to G$, $\psi i'_\alpha = i_\alpha$; uniqueness for G gives $\psi\phi = \mathrm{id}_G$, and $\phi\psi = \mathrm{id}$ as the $i'_\alpha(G_\alpha)$ generate G'. So ϕ is an isomorphism carrying $i_\alpha(G_\alpha)$ onto $i'_\alpha(G_\alpha)$. ∎

Corollary 11.14 (68.6)

If $G = G_1 * G_2$ with $G_1 = \prod^*_{\alpha \in J} H_\alpha$, $G_2 = \prod^*_{\beta \in K} H_\beta$ and $J \cap K = \emptyset$, then $G = \prod^*_{\gamma \in J \cup K} H_\gamma$.

Proof. Homomorphisms h_γ extend to G_1, G_2, then to G, uniquely as the H_γ generate. Apply Lemma 11.13 to the inclusions. ∎

Theorem 11.15 (68.7)

Let $G = G_1 * G_2$, $N_i \lhd G_i$, and N the least normal subgroup of G containing N_1 and N_2. Then $G/N \cong (G_1/N_1) * (G_2/N_2)$.

Proof. Let $F = (G_1/N_1) * (G_2/N_2)$. $G_i \to G_i/N_i \to F$ extend to an onto $\phi : G \to F$ killing N_1, N_2, hence N, inducing $\bar{\phi} : G/N \to F$. $gN_i \mapsto gN$ ($N_i \subset N$) extend to $\psi : F \to G/N$. $\psi\bar{\phi}$ and $\bar{\phi}\psi$ fix the generators gN, gN_i ($g \in G_i$), so are the identity. ∎

Corollary 11.16 (68.8)

If N is the least normal subgroup of $G_1 * G_2$ containing G_1, then $(G_1 * G_2)/N \cong G_2$.

Proof. Take $N_1 = G_1$, $N_2 = \{1\}$ in Theorem 11.15. ■

Lemma 11.17 (68.9)

The least normal subgroup N of G containing $S \subset G$ is generated by the conjugates of elements of S.

Proof. The subgroup N' generated by these conjugates contains S and is normal, as

$$g(x_1 \cdots x_k)g^{-1} = (gx_1g^{-1}) \cdots (gx_kg^{-1})$$

and a conjugate of a conjugate is a conjugate. Every normal subgroup containing S contains N', so $N = N'$. ■

11.3 Free groups

Motivation. Free groups are free products of copies of $\mathbb{Z}$; every group is a quotient of one.

G is a **free group** with **system of free generators** $\{a_\alpha\}$ if each a_α generates an infinite cyclic subgroup G_α and $G = \prod^* G_\alpha$.

Lemma 11.18 (69.1)

If G is free on $\{a_\alpha\}$, then for every group H and elements $y_\alpha \in H$ there is a unique homomorphism $h : G \to H$ with $h(a_\alpha) = y_\alpha$. Conversely, if the a_α generate G and this extension condition holds, G is free on $\{a_\alpha\}$.

Proof. If G is free, $a_\alpha^n \mapsto y_\alpha^n$ defines homomorphisms on the infinite cyclic G_α; extend by Lemma 11.9. Conversely, $H = \mathbb{Z}$, $y_\alpha = 1$ shows G_α is infinite cyclic; homomorphisms $G_\alpha \to H$ correspond to choices of y_α, so the condition of Lemma 11.11 holds for the inclusions, and Lemma 11.13 applies. ■

Theorem 11.19 (69.2)

If $G = G_1 * G_2$ with G_1, G_2 free on $\{a_\alpha\}_{\alpha \in J}$ and $\{a_\alpha\}_{\alpha \in K}$, $J \cap K = \emptyset$, then G is free on $\{a_\alpha\}_{\alpha \in J \cup K}$.

Proof. Corollary 11.14. ■

If G is free on $\{a_\alpha\}$, $\{r_\beta\} \subset G$, and N is the least normal subgroup containing the r_β, a group isomorphic to G/N has the **presentation** with generators a_α and relations r_β. The **commutator subgroup** $[G, G]$ is generated by all $[x, y] = xyx^{-1}y^{-1}$.

Lemma 11.20 (69.3)

$[G, G]$ is normal and $G/[G, G]$ is abelian. Every homomorphism $h : G \to H$ into an abelian group has $[G, G] \subset \ker h$, so it induces $k : G/[G, G] \to H$.

Proof. $g[x,y]g^{-1} = [gxg^{-1}, gyg^{-1}]$; $xy(yx)^{-1} = [x,y]$, so cosets commute; if H is abelian, $h([x,y]) = 1$. ∎

Theorem 11.21 (69.4)

If G is free on $\{a_\alpha\}$, then $G/[G,G]$ is free abelian with basis the cosets $[a_\alpha]$.

Proof. The cosets generate. For y_α in an abelian H, Lemma 11.18 gives $h(a_\alpha) = y_\alpha$, inducing $k([a_\alpha]) = y_\alpha$ (Lemma 11.20). Apply Lemma 11.7. ∎

Corollary 11.22 (69.5)

If G is free on n generators, every system of free generators of G has n elements.

Proof. $G/[G,G]$ is free abelian of rank the number of free generators (Theorem 11.21), unique by Theorem 11.8. (An infinite system would make $G/[G,G]$ not finitely generated.) ∎

11.4 The Seifert–van Kampen theorem

Motivation. $\pi_1(U \cup V)$ is $\pi_1(U) * \pi_1(V)$ with the images of $\pi_1(U \cap V)$ identified.

Theorem 11.23 (70.1, Seifert–van Kampen theorem)

Let $X = U \cup V$ with U, V open and U, V, $U \cap V$ path connected, $x_0 \in U \cap V$. Let i_1, i_2 be induced by the inclusions of $U \cap V$ into U and V, and j_1, j_2 by the inclusions of U and V into X. If $\phi_1 : \pi_1(U, x_0) \to H$ and $\phi_2 : \pi_1(V, x_0) \to H$ are homomorphisms with $\phi_1 \circ i_1 = \phi_2 \circ i_2$, there is a unique homomorphism $\Phi : \pi_1(X, x_0) \to H$ with $\Phi \circ j_1 = \phi_1$ and $\Phi \circ j_2 = \phi_2$.

Proof. Uniqueness: the images of j_1, j_2 generate $\pi_1(X, x_0)$ (Theorem 9.38). Write $[f]_U, [f]_V, [f]$ for path classes in U, V, X.

Step 1. For a loop f at x_0 in U or V let $\rho(f) = \phi_1([f]_U)$ or $\phi_2([f]_V)$ (equal for f in $U \cap V$, as $\phi_1 i_1 = \phi_2 i_2$). Then (1) $[f]_U = [g]_U$ or $[f]_V = [g]_V$ implies $\rho(f) = \rho(g)$;

(2) if f, g both lie in U or both in V, $\rho(f * g) = \rho(f)\rho(g)$.

Step 2. For $x \in X$ choose a path α_x from x_0 to x: constant if $x = x_0$, in $U \cap V$ if $x \in U \cap V$, in U (resp. V) if x lies only in U (resp. V). For a path f in U or V from x to y, $L(f) = \alpha_x * f * \bar{\alpha}_y$ is a loop at x_0 in U or V; let $\sigma(f) = \rho(L(f))$. σ extends ρ ($L(f) \simeq_p f$ for loops at x_0), satisfies (1), and (2) whenever $f * g$ is defined, as $L(f) * L(g) \simeq_p L(f * g)$ in U or V.

Step 3. For a path f in X choose $0 = s_0 < \cdots < s_n = 1$ with each $[s_{i-1}, s_i]$ mapped into U or V, f_i the reparametrized pieces, and $\tau(f) = \sigma(f_1) \cdots \sigma(f_n)$. Adding a point replaces $\sigma(f_i)$ by $\sigma(f_i')\sigma(f_i'')$, equal since $f_i \simeq_p f_i' * f_i''$ in U or V; so τ is well defined (common refinements) and extends σ.

Step 4. If $f \simeq_p g$ in X, then $\tau(f) = \tau(g)$. Special case: a path homotopy F with a subdivision $s_0 < \cdots < s_n$ such that each $R_i = [s_{i-1}, s_i] \times I$ maps into U or V. Let $\beta_i(t) = F(s_i, t)$ (β_0, β_n constant). In R_i the bottom-right and left-top edge paths are path homotopic (convexity), so $f_i * \beta_i \simeq_p \beta_{i-1} * g_i$ in U or V and $\sigma(f_i)\sigma(\beta_i) = \sigma(\beta_{i-1})\sigma(g_i)$. Also $\sigma(\beta_0) = \sigma(\beta_n) = 1$ ($\beta * \beta \simeq_p \beta$ for constant β). Multiplying, $\tau(f) = \sigma(f_1) \cdots \sigma(f_n) = \sigma(g_1) \cdots \sigma(g_n) = \tau(g)$. General case: subdivide $I \times I$ into rectangles $[s_{i-1}, s_i] \times [t_{j-1}, t_j]$ mapped into U or V and apply the special case to consecutive $f_j(s) = F(s, t_j)$.

Step 5. $\tau(f * g) = \tau(f)\tau(g)$: a subdivision for $f * g$ containing $\frac{1}{2}$ splits into subdivisions for f and g.

Step 6. $\Phi([f]) = \tau(f)$ is a well-defined homomorphism (Steps 4, 5), and for a loop f in U, $\Phi(j_1[f]_U) = \tau(f) = \rho(f) = \phi_1([f]_U)$; similarly for V. ■

Theorem 11.24 (70.2, Seifert–van Kampen theorem, classical version)

Under the same hypotheses, let $j : \pi_1(U, x_0) * \pi_1(V, x_0) \to \pi_1(X, x_0)$ extend j_1 and j_2. Then j is surjective, and its kernel is the least normal subgroup N containing all elements $i_1(g)^{-1} i_2(g)$, $g \in \pi_1(U \cap V, x_0)$.

Proof. Surjectivity is Theorem 9.38. As $j_1 i_1 = j_2 i_2$, $N \subset \ker j$, so j induces an epimorphism $k : F/N \to \pi_1(X, x_0)$, F the free product. Let $H = F/N$, ϕ_1, ϕ_2 the inclusions into F followed by projection; $\phi_1 i_1(g) = i_1(g)N = i_2(g)N = \phi_2 i_2(g)$, so Theorem 11.23 gives Φ with $\Phi j_i = \phi_i$. For $g \in \pi_1(U, x_0)$, $\Phi k(gN) = \Phi j_1(g) = gN$, similarly for V; these cosets generate H, so $\Phi \circ k = \mathrm{id}$, k is injective, and $\ker j = N$. ■

Corollary 11.25 (70.3)

If moreover $U \cap V$ is simply connected, then $\pi_1(U, x_0) * \pi_1(V, x_0) \cong \pi_1(X, x_0)$.

Corollary 11.26 (70.4)

If V is simply connected, then $\pi_1(U, x_0)/N \cong \pi_1(X, x_0)$, where N is the least normal subgroup of $\pi_1(U, x_0)$ containing the image of i_1.

Proof. In Theorem 11.24, N is trivial in the first case; in the second, the free product is $\pi_1(U, x_0)$ and $i_1(g)^{-1} i_2(g) = i_1(g)^{-1}$. ■

Example 11.1

A theta space $X = A \cup B \cup C$ (arcs meeting in their end points) has free π_1 of rank 2: for interior points $a \in A$, $b \in B$, $U = X - a$ and $V = X - b$ deformation retract onto the circles $B \cup C$, $A \cup C$, and $U \cap V$ is contractible; apply Corollary 11.25.

11.5 The fundamental group of a wedge of circles

Motivation. A wedge of n circles has π_1 free on n generators.

A Hausdorff space X that is the union of subspaces $S_1, \ldots, S_n$, each homeomorphic to S^1, with $S_i \cap S_j = \{p\}$ for $i \neq j$, is the **wedge** of these circles.

Theorem 11.27 (71.1)

If X is the wedge of $S_1, \ldots, S_n$ and f_i is a loop in S_i representing a generator of $\pi_1(S_i, p)$, then $\pi_1(X, p)$ is free with free generators $[f_1], \ldots, [f_n]$.

Proof. Induction on n; $n = 1$ is clear. Choose $q_i \in S_i - p$, $W_i = S_i - q_i$, $U = S_1 \cup W_2 \cup \cdots \cup W_n$, $V = W_1 \cup S_2 \cup \cdots \cup S_n$; U, V, $U \cap V = W_1 \cup \cdots \cup W_n$ are open and path connected. The open arcs W_i deformation retract onto p, and these fit together (pasting lemma on the closed $W_i \times I$) to a deformation retraction of $U \cap V$ onto p; so $\pi_1(X, p) \cong \pi_1(U, p) * \pi_1(V, p)$ (Corollary 11.25). Similarly U, V deformation retract onto S_1, $S_2 \cup \cdots \cup S_n$, so $\pi_1(U, p) = \langle [f_1] \rangle \cong \mathbb{Z}$ and, by induction, $\pi_1(V, p)$ is free on $[f_2], \ldots, [f_n]$. Apply Theorem 11.19. ■

A union X of subspaces X_α has the topology **coherent** with them if $C \subset X$ is closed whenever each $C \cap X_\alpha$ is closed in X_α. If X is the union of $S_\alpha \cong S^1$ ($\alpha \in J$) with $S_\alpha \cap S_\beta = \{p\}$ for $\alpha \neq \beta$, and coherent with them, X is the **wedge** of the S_α (for finite J this agrees with the earlier definition).

Lemma 11.28 (71.2)

A wedge X of circles S_α is normal, and every compact subspace lies in the union of finitely many S_α.

Proof. Points are closed by coherence. Let A, B be disjoint closed, $p \notin B$. In each S_α choose disjoint open $U_\alpha \supset \{p\} \cup (A \cap S_\alpha)$, $V_\alpha \supset B \cap S_\alpha$. $U = \bigcup U_\alpha$, $V = \bigcup V_\alpha$ are disjoint, $U \cap S_\alpha = U_\alpha$ (all U_β contain p, the only common point), $V \cap S_\alpha = V_\alpha$ ($p \notin V_\beta$), so U, V are open. For compact C choose $x_\alpha \in C \cap (S_\alpha - p)$ where possible; every subset of $D = \{x_\alpha\}$ meets each S_α in at most one point, so is closed; D is closed discrete in the limit point compact C, hence finite. ■

Theorem 11.29 (71.3)

If X is the wedge of circles S_α and f_α represents a generator of $\pi_1(S_\alpha, p)$, then $\pi_1(X, p)$ is free with free generators $\{[f_\alpha]\}$.

Proof. Let G_α be the image of $\pi_1(S_\alpha, p)$. Loops and path homotopies lie in finite unions of circles (Lemma 11.28). Hence, by Theorem 11.27 in finite wedges: the G_α generate; $\pi_1(S_\alpha, p) \to G_\alpha$ is injective, so $G_\alpha \cong \mathbb{Z}$; and a nonempty reduced word representing 1 would already be trivial in a finite wedge, a contradiction. ■

Example 11.2

The **infinite earring** $X \subset \mathbb{R}^2$, the union of the circles C_n of radius $1/n$ centred at $(1/n, 0)$, is not a wedge: its intersection with the positive x-axis meets each C_n in one point but is not closed. Indeed the $[f_n]$ do not generate $\pi_1(X, p)$: let g run through f_n on $[1/(n+1), 1/n]$, $g(0) = p$, a continuous loop. For $N > n$, the retraction of X onto C_N kills $[f_1], \ldots, [f_n]$ but sends $[g]$ to the generator $[f_N]$, so $[g] \notin \langle [f_1], \ldots, [f_n] \rangle$ for every n.

Lemma 11.30 (71.4)

For every index set J there is a wedge of circles S_α, $\alpha \in J$.

Proof. With J discrete, let X be the quotient of $E = S^1 \times J$ collapsing $P = b_0 \times J$ to p, π the quotient map, $S_\alpha = \pi(S^1 \times \alpha)$. For C closed in $S^1 \times \alpha$, $\pi^{-1}\pi(C)$ is C or $C \cup P$, closed, so $\pi(C)$ is closed; hence S_α is closed and $\pi : S^1 \times \alpha \to S_\alpha$ is a homeomorphism. If each $D \cap S_\alpha$ is closed, so is each $\pi^{-1}(D) \cap (S^1 \times \alpha)$, hence $\pi^{-1}(D)$ and D: the topology is coherent. ■

11.6 Adjoining a two-cell

Motivation. Attaching a disc along a loop adds that loop as a relation.

Theorem 11.31 (72.1)

Let X be Hausdorff, $A \subset X$ closed and path connected, and $h : B^2 \to X$ continuous, mapping Int B^2 bijectively onto $X - A$ and S^1 into A. Let $p \in S^1$, $a = h(p)$ and $k = h|S^1 : (S^1, p) \to$

(A, a). Then $i_* : \pi_1(A, a) \to \pi_1(X, a)$ induced by inclusion is surjective, and its kernel is the least normal subgroup containing $k_*(\pi_1(S^1, p))$.

Proof.

Step 1. With $x_0 = h(0)$, A is a deformation retract of $U = X - x_0$. $C = h(B^2)$ is closed; $\pi = h : B^2 \to C$ gives a closed map $\pi \times \mathrm{id} : B^2 \times I \to C \times I$ (compact to Hausdorff), a quotient map, as is its restriction to the saturated open $(B^2 - 0) \times I$. So the radial deformation retraction of $B^2 - 0$ onto S^1 induces one of $C - x_0$ onto $\pi(S^1)$; extend it by fixing A (pasting lemma: the closed sets $C - x_0$ and A of U meet in the fixed $\pi(S^1)$). So $\pi_1(A, a) \cong \pi_1(U, a)$ (Theorem 9.33), and it suffices that $\pi_1(U, a) \to \pi_1(X, a)$ is onto with kernel the least normal subgroup containing $[g]$, $g = h \circ f$, f a generator of $\pi_1(S^1, p)$.

Step 2. $V = X - A = h(\operatorname{Int} B^2)$ is open; $\pi | \operatorname{Int} B^2$ is a bijective quotient map (saturated open restriction), a homeomorphism, so V is simply connected. $U \cap V = V - x_0 \cong \operatorname{Int} B^2 - 0$ is path connected with $\pi_1 \cong \mathbb{Z}$, and U is path connected. For $b \in U \cap V$, Corollary 11.26: $\pi_1(U, b) \to \pi_1(X, b)$ is onto with kernel the least normal subgroup containing the image of $\pi_1(U \cap V, b)$.

Step 3. Let q be the midpoint of $0p$, $b = h(q)$, f_0 a generating loop of $\operatorname{Int} B^2 - 0$ at q, $g_0 = h \circ f_0$, γ the segment from q to p, $\delta = h \circ \gamma$. The $\hat{\delta}$ commute with inclusion $U \to X$, so $\pi_1(U, a) \to \pi_1(X, a)$ is onto with kernel the least normal subgroup containing $\hat{\delta}([g_0])$. $\bar{\gamma} * f_0 * \gamma$ generates $\pi_1(B^2 - 0, p)$, so is path homotopic there to f or $\bar{f}$; applying h, $\hat{\delta}([g_0]) = [g]^{\pm 1}$ in $\pi_1(U, a)$. ■

A space homeomorphic to B^2 is a **2-cell**; X is obtained from A by **adjoining a 2-cell**.

11.7 The fundamental groups of the torus and the dunce cap

Motivation. Polygon models give $\pi_1(T) = \langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2$ and π_1 of the dunce cap.

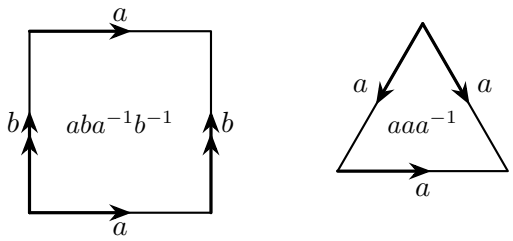


Figure. *Left: the torus, boundary word* $aba^{-1}b^{-1}$, $\pi_1 \cong \mathbb{Z} \times \mathbb{Z}$. *Right: a dunce cap, boundary word* aaa^{-1}, $\pi_1 = \langle a \mid a^2a^{-1} \rangle$ *trivial.*

Theorem 11.32 (73.1)

The fundamental group of the torus has a presentation with two generators α, β and one relation $\alpha\beta\alpha^{-1}\beta^{-1}$.

Proof. Let $h : I^2 \to T = S^1 \times S^1$ be the restriction of $p \times p$, $A = h(\operatorname{Bd} I^2)$, a wedge of two circles, $a = h(0, 0)$; Theorem 11.31 applies ($I^2 \cong B^2$). With $a_0(t) = (t, 0)$, $b_0(t) = (0, t)$, $a_1(t) = (t, 1)$, $b_1(t) = (1, t)$, $\alpha = h \circ a_0$, $\beta = h \circ b_0$ freely generate $\pi_1(A, a)$ (Theorem 11.27). $f = a_0 * (b_1 * (\bar{a}_1 * \bar{b}_0))$ generates $\pi_1(\operatorname{Bd} I^2, p)$, and $h \circ f = \alpha * (\beta * (\bar{\alpha} * \bar{\beta}))$. Apply Theorem 11.31. ■

Corollary 11.33 (73.2)

The fundamental group of the torus is free abelian of rank 2.

Proof. Let G be free on α, β, N the least normal subgroup containing the commutator $\alpha\beta\alpha^{-1}\beta^{-1}$; so $N \subset [G,G]$. G/N is generated by the commuting $\alpha N, \beta N$, so abelian and $[G,G] \subset N$. Hence $G/N = G/[G,G]$, free abelian of rank 2 (Theorem 11.21). ■

For $n > 1$ let r be rotation of S^1 by $2\pi/n$. The **n-fold dunce cap** X is the quotient of B^2 identifying each $x \in S^1$ with $r(x), \dots, r^{n-1}(x)$. The quotient map π is closed: $\pi^{-1}\pi(C) = C \cup r(C_0) \cup \cdots \cup r^{n-1}(C_0)$, $C_0 = C \cap S^1$. So X is compact, and Hausdorff by the next lemma.

Lemma 11.34 (73.3)

If $\pi : E \to X$ is a closed quotient map and E is normal, so is X.

Proof. Points of X are closed. For disjoint closed $A, B \subset X$ take disjoint open $U \supset \pi^{-1}(A)$, $V \supset \pi^{-1}(B)$. $U_0 = X - \pi(E - U)$ and $V_0 = X - \pi(E - V)$ are open (π closed), contain A, B, and are disjoint: $x \in U_0 \Rightarrow \pi^{-1}(x) \subset U$, $x \in V_0 \Rightarrow \pi^{-1}(x) \subset V$. ■

The 2-fold dunce cap is P^2: $B^2 \to$ upper hemisphere $\to P^2$ is a closed continuous surjection, injective on $\operatorname{Int} B^2$, identifying x with $-x$ on S^1.

Theorem 11.35 (73.4)

The fundamental group of the n-fold dunce cap is cyclic of order n.

Proof. Let $h : B^2 \to X$ be the quotient map, $A = h(S^1)$, $p = (1,0)$, $a = h(p)$. h maps the arc of S^1 from p to $r(p)$ onto A, identifying only its end points, so A is a simple closed curve and, for $\gamma(t) = (\cos 2\pi t/n, \sin 2\pi t/n)$, $\alpha = h \circ \gamma$ generates $\pi_1(A,a) \cong \mathbb{Z}$. $f = \gamma * (r \circ \gamma) * \cdots * (r^{n-1} \circ \gamma)$ generates $\pi_1(S^1, p)$, and as $h \circ r^m = h$ on S^1, $h \circ f = \alpha^n$. By Theorem 11.31, $\pi_1(X,a) \cong \mathbb{Z}/n\mathbb{Z}$. ■